

N.S.LABORATORY SDN. BHD.

T-MULIN 20, 200g/kg, Premix for medicated feeding stuff

Active Ingredient:

Each kg contains

Tiamulin (as hydrogen fumarate) ----- 200g

Product Description:

White or almost white powder which is incorporated in animal feeding stuff

Dosage form:

Premix for medicated feeding stuff

Pharmacodynamics:

Tiamulin is a bacteriostatic semi-synthetic antibiotic belonging to the pleuromutilin group of antibiotics and acts at the ribosomal level to inhibit bacterial protein synthesis.

Tiamulin has shown in-vitro activity against a wide range of bacteria including *Brachyspira hyodysenteriae*, *Brachyspira pilosicoli*, *Lawsonia intracellularis* and *Mycoplasma* spp.

Tiamulin is bacteriostatic at therapeutic concentrations and has been shown to act at the 70S ribosome level and the primary binding site is on the 50S subunit and possibly a secondary site where the 50S and 30S subunits join. It appears to inhibit microbial protein production by producing biochemical inactive initiation complexes, which prevent elongation of the polypeptide chain.

Mechanisms responsible for resistance development in *Brachyspira* spp to the pleuromutilin class of antibiotics are considered to be based on mutations at the ribosomal target site. Clinically relevant resistance to tiamulin requires combinations of mutations around the tiamulin binding site. Resistance to tiamulin may be associated with decreased susceptibility to other pleuromutilins.

Pharmacokinetics:

Pigs- Following oral administration, tiamulin hydrogen fumarate is rapidly absorbed from the gastrointestinal tract of pigs (85-90%) and appears in the blood within 30 minutes. 2-4 hours (t_{max}) after the oral administration of 10 mg tiamulin/kg BW in the form of an oral solution, a C_{max} of 1 µg/ml was measured; an oral administration of 25 mg/kg gave a C_{max} of 1.82 µg/ml.

There is very good distribution in the tissues with accumulation in lungs and in the colon. 30-50% of tiamulin is bound to serum proteins. Tiamulin is rapidly metabolised in the liver (hydroxylation, de-alkalisation, hydrolysis). At least 16 biologically inactive metabolites have been identified. The excretion of tiamulin and its metabolites is through the bile and faeces (70-85%). The remainder is excreted through the urine (15-30%).

Chickens- Tiamulin is well absorbed in chickens (70-95%) after oral administration.

Tiamulin distributes widely through the body and has been shown to concentrate in the liver and kidney (sites of excretion) and in the lung (30 times serum level). Excretion is mainly via the bile (55-65%) and kidney (15-30%) as mainly microbiologically inactive metabolites and is quite rapid, 99% of the dose within 48 hours.

Turkeys- In turkeys serum levels of tiamulin are similar to chickens. In breeders on 0.025% tiamulin the average serum level was 0.36µg/ml (range 0.22-0.5µg/ml).

Rabbits- There are no pharmacokinetic data available for rabbits.

Indication:

Pigs- For the treatment and metaphylaxis, when the disease is present in the group, of swine dysentery caused by *Brachyspira hyodysenteriae* susceptible to tiamulin. The presence of the disease in the group must be established before the product is used. For the treatment of colitis caused by *Brachyspira pilosicoli*. For the treatment of ileitis caused by *Lawsonia intracellularis*. For the treatment of enzootic pneumonia caused by *Mycoplasma hyopneumoniae*.

Chickens- For the treatment and metaphylaxis, when the disease is present at herd level, of chronic respiratory disease (CRD) and airsacculitis caused by *Mycoplasma gallisepticum* and *Mycoplasma synoviae* susceptible to tiamulin. The presence of the disease in the herd should be established before use.

Turkeys- For the treatment and metaphylaxis, when the disease is present at herd level, of infectious sinusitis and airsacculitis caused by *Mycoplasma gallisepticum*, *Mycoplasma meleagridis* and *Mycoplasma synoviae* susceptible to tiamulin. The presence of the disease in the herd should be established before use.

Rabbits- For the treatment and metaphylaxis, when the disease is present at herd level, of epizootic rabbit enterocolitis (ERE) caused by pathogens susceptible to tiamulin. The presence of the disease in the herd should be established before use.

Mode of Administration:

Oral administration after incorporation in animal feed

Recommended Dosage:

For oral administration only after incorporation in medicated feed.

The uptake of medicated feed depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of tiamulin should be adjusted using the following formula:

$$kg \text{ premix/tonne} = \frac{\text{Dose rate}(mg/kg) \times \text{mean bodyweight}(kg)}{\text{Mean feed intake}(kg) \times \text{premix strength}(mg/g)}$$

To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.

Pigs- Treatment and metaphylaxis of Swine Dysentery caused by *B. hyodysenteriae*, treatment of Porcine Colonic Spirochaetosis (colitis) caused by *B. pilosicoli*.

Dosage: 5 – 10 mg tiamulin hydrogen fumarate (equivalent to 4.05 – 8.1 mg tiamulin base) / kg bodyweight daily administered for 7 to 10 consecutive days. The dosage will normally be achieved by an inclusion level of 100 – 200 ppm tiamulin hydrogen fumarate in the finished feed providing that feed intake is unaffected.

Treatment of Porcine Proliferative Enteropathy (ileitis) caused by *L. intracellularis*
Dosage: 7.5 mg tiamulin hydrogen fumarate (equivalent to 6.075 mg tiamulin base) / kg bodyweight daily administered for 10-14 consecutive days. The dosage will normally be achieved by an inclusion level of 150 ppm tiamulin hydrogen fumarate in the finished feed providing that feed intake is unaffected.

Treatment of Enzootic Pneumonia caused by *M. hyopneumoniae*
Dosage: 5.0 – 10.0 mg tiamulin hydrogen fumarate (equivalent to 4.05 – 8.1 mg tiamulin base) / kg bodyweight daily administered for 7 to 10 consecutive days. The dosage will normally be achieved by an inclusion level of 100 - 200 ppm tiamulin hydrogen fumarate in the finished feed, providing that feed intake is unaffected.

Secondary infection by organisms such as *Pasteurella multocida* and *Actinobacillus pleuropneumoniae* may complicate enzootic pneumonia and require specific medication.

Chickens (broiler, replacement pullet, laying and breeding hens)- Treatment and metaphylaxis of chronic respiratory disease (CRD) caused by *M. gallisepticum* and airsacculitis and infectious synovitis caused by *M. synoviae*.

Dosage - Treatment and metaphylaxis: 25 mg tiamulin hydrogen fumarate (equivalent to 20.25 mg tiamulin base) / kg body weight daily administered for the period of 3 to 5 consecutive days. This is normally achieved by an inclusion level of 250 - 500 ppm tiamulin hydrogen fumarate in the finished feed provided that feed intake is unaffected.

Turkeys (young poults, breeding turkeys)- Treatment and metaphylaxis of infectious sinusitis and airsacculitis caused by *M. gallisepticum*, *M. synoviae* and *M. meleagridis*.

Dosage - Treatment and metaphylaxis: 40 mg tiamulin hydrogen fumarate (equivalent to 32.4 mg tiamulin base) / kg body weight daily administered for the period of 3 to 5 consecutive days. This is normally achieved by an inclusion level of 250 - 500 ppm tiamulin hydrogen fumarate in finished feed provided that feed intake is unaffected.

Metaphylaxis with tiamulin should only be initiated after confirmed infection with *M. gallisepticum*, *M. synoviae* and *M. meleagridis* and then as an aid in the metaphylaxis strategy to reduce the clinical signs and mortality from respiratory disease in flocks, where infection in ovum is likely because the disease is known to exist in the parent generation. The metaphylaxis strategy should include efforts to eliminate the infection from the parent generation.

Rabbits- Treatment of Epizootic Rabbit Enterocolitis (ERE) and metaphylaxis of ERE in farms with clinical signs of ERE in the previous fattening cycle as part of a programme including measures aiming to eradicate or control the infection in the farm.

Dosage: 3 mg tiamulin hydrogen fumarate (equivalent to 2.43 mg tiamulin base) / kg body weight daily. The dosage will normally be achieved by an inclusion level of 40 ppm tiamulin hydrogen fumarate in the finished feed provided that feed intake is unaffected. Treatment should be administered until 2 - 3 days after clinical signs have resolved. Metaphylaxis should be administered during 3 - 4 weeks from the first week after weaning.

Contraindications:

Do not use in cases of known hypersensitivity to the active substances or any of the excipients.

Do not use in case of resistance to tiamulin.

Do not administer products containing ionophores such as monensin, salinomycin or narasin during or for at least seven days before or after treatment with the product. Severe growth depression or death may result

Special Warnings for each Target Species:

The uptake of medication by animals can be altered as a consequence of illness. For animals with a reduced feed intake, treat parenterally using an appropriate injectable product.

Long term or repeated use should be avoided by improving management practice and thorough cleansing and disinfection.

In case of reduced feed intake, the inclusion levels in feed may need to be increased to achieve target dosage.

Special Precautions for Use in Animals:

Do not use the product in liquid feed.

Due to the likely variability (time, geographical) in the occurrence of resistance of bacteria for tiamulin, the use of the product should be based on susceptibility testing and take into account official and local antimicrobial policies. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to tiamulin and may decrease the effectiveness of treatment with other pleuromutilins due to the potential for crossresistance. If there is no response to treatment within 3 days, the diagnosis should be re-established.

Inform the feed supplier that tiamulin will be used, to avoid incorporating of ionophore products containing monensin, narasin and salinomycin products in the feed and to avoid contamination of the feed. In case of a suspected contamination, test the feed for the presence of these ionophores before feeding. If adverse effects occur due to an interaction, stop administration of the feed immediately. Remove the contaminated feed as soon as possible and replace with uncontaminated feed.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:

Direct contact with the skin, eyes and mucous membranes should be avoided by wearing overalls, impermeable rubber gloves and safety glasses when mixing or handling the product.

In case of accidental eye contact, irrigate the eyes thoroughly with clean running water immediately. Seek medical advice if irritation persists.

When handling the product, inhalation of the dust must be avoided by wearing a mask. Contaminated clothing should be removed and any splashes on to the skin should be washed off immediately.

Wash hands after use.

Accidental ingestion should be avoided. In case of accidental ingestion, seek medical advice immediately and show the package leaflet or label to the physician.

People with known hypersensitivity to tiamulin should avoid contact with the veterinary medicinal product.

Interactions with other Medicaments:

Tiamulin is known to produce clinically important (often lethal) interactions with ionophore antibiotics, including monensin, narasin, salinomycin. Therefore, animals should not receive products containing such compounds during or for at least seven days before or after treatment with this product. Severe growth depression, ataxia, paralysis or death may result. Tiamulin may lessen the antibacterial activity of beta-lactam antibiotics, whose action is dependent on bacterial growth.

Usage during Pregnancy and Lactation:

Can be used in pigs during pregnancy and lactation.

Can be used in laying and breeding chickens and turkeys.

Can be used in rabbits during pregnancy and lactation.

Adverse Effects:

In rare cases, hypersensitivity to tiamulin following oral administration to pigs is reported in terms of acute dermatitis with cutaneous erythema and intense pruritus. The adverse reactions are usually mild and transient but in very rare cases may be serious. If these typical side effects occur, stop treatment immediately and clean animals and pens with water. Normally, affected animals recover quickly. Symptomatic treatment such as electrolyte therapy and an anti-inflammatory therapy may be useful.

Overdose and Treatment:

Pigs: A single oral dose of 100 mg/kg BW caused hyperpnoea and abdominal discomfort in pigs. At a dose of 150 mg/kg the only effect on the central nervous system was lethargy. A dose of 55 mg/kg for 14 days caused increased salivation and a mild irritation of the stomach. Tiamulin hydrogen fumarate has a relatively high therapeutic index in pigs. The minimum lethal dose has not been established in pigs.

Chickens and turkeys: The LD₅₀ for chickens is 1290mg/kg and turkeys 840 mg/kg bodyweight. The clinical signs of acute toxicity in chickens are - vocalization, clonic cramps and lateral recumbency. In turkeys signs of acute toxicity include clonic cramps, lateral or dorsal recumbency, salivation and ptosis.

If signs of intoxication do occur promptly remove the medicated feed, replace with fresh unmedicated feed and apply supportive, symptomatic therapy.

Withdrawal Periods (Meat and Offal):

Pigs- 6 days

Chickens- 1 day, Eggs: 0 days

Turkeys- 4 days

Rabbits- 0 days

Storage Condition:

Store in cool, dark place(< 30°C). Protect from moisture and direct sunlight.

Shelf Life:

36 months

Shelf Life after Opening:

Discard any unused powder immediately after opening

Shelf Life after Incorporation into Feed:

24 hours

Packaging Available:

500g/packet, 30packets/carton, 15kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

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